

# **Application of R for Finance - Group Assignment 2**

Group 28

2025-10-26

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|----|-----------------------------------------|----|

```
library(dplyr)
library(lubridate)
library(ggplot2)
```

## Loading and Preparing the Data

We import the raw Compustat file and remove the NA values

```
#remove NA values to prepare the data
sp500 <- sp500 %>%
  filter(!is.na(prcod)) %>%
  select( -gvkey)

head(sp500)
```

A data.frame: 6 × 10

|   | tic<br><chr> | data-<br>date<br><date> | conm<br><chr>                                 | exchg<br><int> | sic<br><int> | cshtd<br><dbl> | prccd<br><dbl> | prchd<br><dbl> | prcld<br><dbl> | prcod<br><dbl> |
|---|--------------|-------------------------|-----------------------------------------------|----------------|--------------|----------------|----------------|----------------|----------------|----------------|
| 1 | A            | 2023-01-03              | AGI-<br>LENT<br>TECH-<br>NOLO-<br>GIES<br>INC | 11             | 3826         | 1414284        | 150.04         | 153.13         | 148.470        | 151.96         |
| 2 | A            | 2023-01-04              | AGI-<br>LENT<br>TECH-<br>NOLO-<br>GIES<br>INC | 11             | 3826         | 1247420        | 151.67         | 153.04         | 150.240        | 151.65         |
| 3 | A            | 2023-01-05              | AGI-<br>LENT<br>TECH-<br>NOLO-<br>GIES<br>INC | 11             | 3826         | 1714635        | 152.11         | 153.07         | 148.770        | 150.00         |

|   | tic<br><chr> | data-<br>date<br><date> | conm<br><chr>                                 | exchg<br><int> | sic<br><int> | cshtd<br><dbl> | prccd<br><dbl> | prchd<br><dbl> | prcld<br><dbl> | prcod<br><dbl> |
|---|--------------|-------------------------|-----------------------------------------------|----------------|--------------|----------------|----------------|----------------|----------------|----------------|
| 4 | A            | 2023-01-06              | AGI-<br>LENT<br>TECH-<br>NOLO-<br>GIES<br>INC | 11             | 3826         | 2444397        | 147.67         | 154.64         | 143.010        | 154.36         |
| 5 | A            | 2023-01-09              | AGI-<br>LENT<br>TECH-<br>NOLO-<br>GIES<br>INC | 11             | 3826         | 1269488        | 147.47         | 151.28         | 147.198        | 149.69         |
| 6 | A            | 2023-01-10              | AGI-<br>LENT<br>TECH-<br>NOLO-<br>GIES<br>INC | 11             | 3826         | 1565745        | 155.23         | 155.55         | 148.750        | 150.18         |

```
# Make sure Date is properly formatted
colnames(sp500)

sp500 <- sp500 %>%
  mutate(datadate = dmy(datadate))

head(sp500)
```

1. 'tic'
2. 'datadate'
3. 'conm'
4. 'exchg'
5. 'sic'
6. 'cshtd'
7. 'prccd'
8. 'prchd'
9. 'prcld'
10. 'prcod'
11. 'gvkey'

A data.frame: 6 × 11

|   | tic<br><chr> | data-<br>date<br><date> | conm<br><chr>                          | ex-<br>chg<br><int> | sic<br><int> | cshtd<br><dbl> | prccd<br><dbl> | prchd<br><dbl> | prcld<br><dbl> | prcod<br><dbl> | gvkey<br><int> |
|---|--------------|-------------------------|----------------------------------------|---------------------|--------------|----------------|----------------|----------------|----------------|----------------|----------------|
| 1 | PNW          | 2023-01-03              | PIN-NA-CLE<br>WEST<br>CAP-ITAL<br>CORP | 11                  | 4911         | 1442534        | 74.63          | 76.4125        | 73.380         | 76.25          | 1075           |
| 2 | PNW          | 2023-01-04              | PIN-NA-CLE<br>WEST<br>CAP-ITAL<br>CORP | 11                  | 4911         | 954218         | 75.39          | 76.0950        | 74.630         | 75.10          | 1075           |
| 3 | PNW          | 2023-01-05              | PIN-NA-CLE<br>WEST<br>CAP-ITAL<br>CORP | 11                  | 4911         | 994775         | 73.65          | 75.0950        | 73.305         | 74.88          | 1075           |
| 4 | PNW          | 2023-01-06              | PIN-NA-CLE<br>WEST<br>CAP-ITAL<br>CORP | 11                  | 4911         | 729808         | 75.46          | 76.0200        | 74.480         | 74.49          | 1075           |
| 5 | PNW          | 2023-01-09              | PIN-NA-CLE<br>WEST<br>CAP-ITAL<br>CORP | 11                  | 4911         | 656127         | 75.55          | 76.4800        | 75.240         | 75.24          | 1075           |

|   | tic<br><chr> | data-<br>date<br><date> | conm<br><chr>              | ex-<br>chg<br><int> | sic<br><int> | cshtd<br><dbl> | prccd<br><dbl> | prchd<br><dbl> | prcld<br><dbl> | prcod<br><dbl> | gvkey<br><int> |
|---|--------------|-------------------------|----------------------------|---------------------|--------------|----------------|----------------|----------------|----------------|----------------|----------------|
| 6 | PNW          | 2023-01-10              | PIN-NA-CLEWESTCAP-ITALCORP | 11                  | 4911         | 763254         | 75.65          | 75.6950        | 74.880         | 75.31          | 1075           |

## Part I: Q&A

Interrogate the data to answer the questions in A2 Questions.pdf

### 1. How many unique tickers are in your data?

```
# Number of unique tickers in the data
no_unique_tickers <- sp500 %>%
  summarise(n_tickers = n_distinct(tic))
cat("There are 502 unique tickers in the data", "\n\n")
```

There are 502 unique tickers in the data

### 2. How many unique companies are in your data?

```
no_unique_companies <- sp500 %>%
  summarise(n_companies = n_distinct(conm))
cat("There are 499 unique companies in the data", "\n\n")
```

There are 499 unique companies in the data

### 3. Display the top 5 companies by largest mean trading volume, in a table.

```
#Displaying top 5 companies by largest mean trading volume
top_5_volume <- sp500 %>%
  group_by(conm) %>%
  summarise(mean_volume = mean(cshtrd, na.rm = TRUE)) %>%
  arrange(desc(mean_volume)) %>%
  slice(1:5)
top_5_volume
```

A tibble: 5 × 2

Table 3: Top 5 companies by largest mean trading volume

| conm <chr>             | mean__volume <dbl> |
|------------------------|--------------------|
| TESLA INC              | 115314383          |
| NVIDIA CORP            | 113131835          |
| PALANTIR TECHNOLOG INC | 60056251           |
| APPLE INC              | 57736403           |
| ADVANCED MICRO DEVICES | 57143415           |

#### 4. Display the total trading volume of the top 3 exchanges by largest total trading volume, in a table.

```
# Calculate the total trading volume by exchange
#|tbl-cap: "Total trading volume of top 3 exchanges by largest trading volume"
top_3_exchanges <- sp500 %>%
  group_by(exchg) %>%
  summarise(total_volume = sum(cshtrd, na.rm = TRUE)) %>%
  arrange(desc(total_volume)) %>%
  slice(1:3)
top_3_exchanges
```

A tibble: 3 × 2

| exchg <int> | total__volume <dbl> |
|-------------|---------------------|
| 11          | 681415756062        |
| 14          | 570830885382        |
| 21          | 385399362           |



**5. Visualise the total trading volume of the top 3 exchanges by largest total trading volume, in a bar plot.**

```
# Order bars largest → smallest
top_3_exchanges <- top_3_exchanges %>%
  arrange(desc(total_volume)) %>%
  mutate(exchg = factor(exchg, levels = exchg))

# Create colors for each exchange
cols <- c("tomato", "steelblue", "skyblue3") # distinct but cohesive palette

ggplot(top_3_exchanges, aes(x = exchg, y = total_volume, fill = exchg)) +
  geom_col(show.legend = TRUE) +
  scale_fill_manual(
    values = cols,
    name = "Exchange", # legend title
    labels = function(x) paste("Exchange:", x) # optional, adds clarity
  ) +
  labs(
    x = "Exchange",
    y = "Total Trading Volume",
    title = "Top 3 Exchanges by Total Trading Volume"
  ) +
  scale_y_continuous(
    labels = function(x) format(x, big.mark = ",", scientific = FALSE),
    limits = c(0, max(top_3_exchanges$total_volume) * 1.05),
    expand = c(0, 0)
  ) +
  theme_minimal(base_family = "Arial") +
  theme(
    legend.position = "right", # move legend to right of chart
    legend.title = element_text(size = 11, face = "bold"),
    legend.text = element_text(size = 10),
    plot.title = element_text(size = 13, face = "bold", hjust = 0.5),
    axis.title = element_text(size = 11),
    axis.text = element_text(size = 10)
  )
```

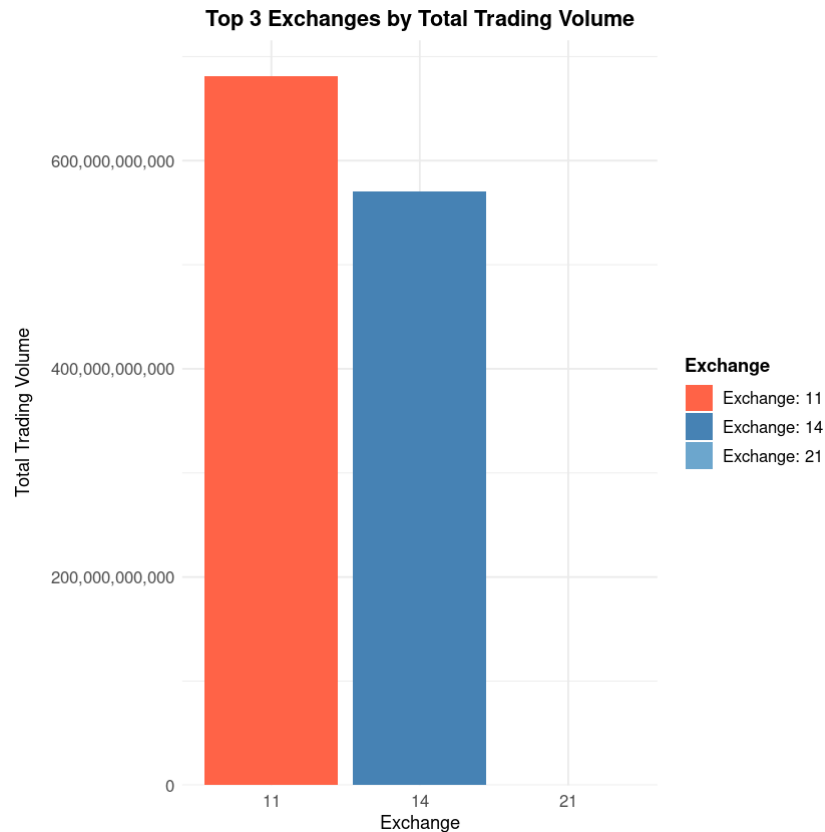


Figure 1: Visualisation of the total trading volume of the top 3 exchanges

## 6. How many companies have more than one ticker?

```
# Make a table of companies with >1 ticker
multi_tic_companies <- sp500 %>%
  group_by(conm) %>%
  summarise(n_tickers = n_distinct(tic)) %>%
  filter(n_tickers > 1)

# Count how many such companies there are
cat("There are 3 companies with more than one ticker in the data", "\n\n")
```

There are 3 companies with more than one ticker in the data

## 7. Which ticker has the largest positive mean return (simple daily return)?

```
# Simple daily return
returns_by_tic <- sp500 %>%
  arrange(tic, datadate) %>%
  group_by(tic) %>%
  mutate(ret = prccd / lag(prccd) - 1) %>%
  ungroup()

ticker_largest_pos <- returns_by_tic %>%
  filter(!is.na(ret)) %>%
  group_by(tic) %>%
  summarise(mean_return = mean(ret, na.rm = TRUE)) %>%
  filter(mean_return > 0) %>%
  arrange(desc(mean_return)) %>%
  slice(1)

ticker_largest_pos
cat("PLTR has the largest positive mean return, (simple daily return)", "\n\n")
```

A tibble: 1 × 2

Table 5: Ticker with the largest positive mean return

| tic <chr> | mean_return <dbl> |
|-----------|-------------------|
| PLTR      | 0.005785119       |

PLTR has the largest positive mean return, (simple daily return)

## 8. Which company has the largest positive mean return (simple daily return)?

```
#Which company has the largest positive mean return
company_largest_pos <- returns_by_tic %>%
  filter(!is.na(ret)) %>%
  group_by(comm) %>%
  summarise(mean_return = mean(ret, na.rm = TRUE)) %>%
  filter(mean_return > 0) %>%
  arrange(desc(mean_return)) %>%
  slice(1)
```

```
company_largest_pos
cat("Largest Positive Mean Return = Palantir Technologies Inc", "\n\n")
```

A tibble: 1 × 2

Table 6: Company with the largest positive mean return

| conm <chr>             | mean_return <dbl> |
|------------------------|-------------------|
| PALANTIR TECHNOLOG INC | 0.005785119       |

Largest Positive Mean Return = Palantir Technologies Inc

## 9. Which industry is represented by the most companies?

```
#Which Industry is represented by the most companies
top_industry <- sp500 %>%
  group_by(sic) %>%
  summarise(n_companies = n_distinct(conm)) %>%
  arrange(desc(n_companies)) %>%
  slice(1)

top_industry
```

A tibble: 1 × 2

Table 7: Industry represented by the most companies

| sic <int> | n_companies <int> |
|-----------|-------------------|
| 6798      | 28                |

## Loading and Preparing the Data

We import the raw Compustat file and remove the NA values

```
sp500 <- read.csv("sp500_2023_2024.csv")
```

```
#remove NA values to prepare the data
sp500 <- sp500 %>%
  filter(!is.na(prcod))

head(sp500)
```

A data.frame: 6 × 10

|   | tic<br><chr> | data-<br>date<br><date> | conm<br><chr>                                 | exchg<br><int> | sic<br><int> | cshtd<br><dbl> | prccd<br><dbl> | prchd<br><dbl> | prcld<br><dbl> | prcod<br><dbl> |
|---|--------------|-------------------------|-----------------------------------------------|----------------|--------------|----------------|----------------|----------------|----------------|----------------|
| 1 | A            | 2023-01-03              | AGI-<br>LENT<br>TECH-<br>NOLO-<br>GIES<br>INC | 11             | 3826         | 1414284        | 150.04         | 153.13         | 148.470        | 151.96         |
| 2 | A            | 2023-01-04              | AGI-<br>LENT<br>TECH-<br>NOLO-<br>GIES<br>INC | 11             | 3826         | 1247420        | 151.67         | 153.04         | 150.240        | 151.65         |
| 3 | A            | 2023-01-05              | AGI-<br>LENT<br>TECH-<br>NOLO-<br>GIES<br>INC | 11             | 3826         | 1714635        | 152.11         | 153.07         | 148.770        | 150.00         |
| 4 | A            | 2023-01-06              | AGI-<br>LENT<br>TECH-<br>NOLO-<br>GIES<br>INC | 11             | 3826         | 2444397        | 147.67         | 154.64         | 143.010        | 154.36         |
| 5 | A            | 2023-01-09              | AGI-<br>LENT<br>TECH-<br>NOLO-<br>GIES<br>INC | 11             | 3826         | 1269488        | 147.47         | 151.28         | 147.198        | 149.69         |

|   | tic   | data-<br>date | comm                                          | exchg | sic   | cshtd   | prccd  | prchd  | prcld   | prcod  |
|---|-------|---------------|-----------------------------------------------|-------|-------|---------|--------|--------|---------|--------|
|   | <chr> | <date>        | <chr>                                         | <int> | <int> | <dbl>   | <dbl>  | <dbl>  | <dbl>   | <dbl>  |
| 6 | A     | 2023-01-10    | AGI-<br>LENT<br>TECH-<br>NOLO-<br>GIES<br>INC | 11    | 3826  | 1565745 | 155.23 | 155.55 | 148.750 | 150.18 |

```
# Make sure Date is properly formatted
colnames(sp500)
```

```
sp500 <- sp500 %>%
  mutate(datadate = dmy(datadate))
```

```
head(sp500)
```

1. 'tic'
2. 'datadate'
3. 'comm'
4. 'exchg'
5. 'sic'
6. 'cshtd'
7. 'prccd'
8. 'prchd'
9. 'prcld'
10. 'prcod'
11. 'gvkey'

A data.frame: 6 × 11

|   | tic<br><chr> | data-<br>date<br><date> | conm<br><chr>                          | ex-<br>chg<br><int> | sic<br><int> | cshtd<br><dbl> | prccd<br><dbl> | prchd<br><dbl> | prcld<br><dbl> | prcod<br><dbl> | gvkey<br><int> |
|---|--------------|-------------------------|----------------------------------------|---------------------|--------------|----------------|----------------|----------------|----------------|----------------|----------------|
| 1 | PNW          | 2023-01-03              | PIN-NA-CLE<br>WEST<br>CAP-ITAL<br>CORP | 11                  | 4911         | 1442534        | 74.63          | 76.4125        | 73.380         | 76.25          | 1075           |
| 2 | PNW          | 2023-01-04              | PIN-NA-CLE<br>WEST<br>CAP-ITAL<br>CORP | 11                  | 4911         | 954218         | 75.39          | 76.0950        | 74.630         | 75.10          | 1075           |
| 3 | PNW          | 2023-01-05              | PIN-NA-CLE<br>WEST<br>CAP-ITAL<br>CORP | 11                  | 4911         | 994775         | 73.65          | 75.0950        | 73.305         | 74.88          | 1075           |
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| 5 | PNW          | 2023-01-09              | PIN-NA-CLE<br>WEST<br>CAP-ITAL<br>CORP | 11                  | 4911         | 656127         | 75.55          | 76.4800        | 75.240         | 75.24          | 1075           |

|   | tic<br><chr> | data-<br>date<br><date> | conm<br><chr>                          | ex-<br>chg<br><int> | sic<br><int> | cshtd<br><dbl> | prccd<br><dbl> | prchd<br><dbl> | prcld<br><dbl> | prcod<br><dbl> | gvkey<br><int> |
|---|--------------|-------------------------|----------------------------------------|---------------------|--------------|----------------|----------------|----------------|----------------|----------------|----------------|
| 6 | PNW          | 2023-01-10              | PIN-NA-CLE<br>WEST<br>CAP-ITAL<br>CORP | 11                  | 4911         | 763254         | 75.65          | 75.6950        | 74.880         | 75.31          | 1075           |

```
# Sort data by Ticker and Date
sp500 <- sp500 %>%
  arrange(tic, datadate)

head(sp500)
```

A data.frame: 6 × 10

|   | tic<br><chr> | data-<br>date<br><date> | conm<br><chr>                         | exchg<br><int> | sic<br><int> | cshtd<br><dbl> | prccd<br><dbl> | prchd<br><dbl> | prcld<br><dbl> | prcod<br><dbl> |
|---|--------------|-------------------------|---------------------------------------|----------------|--------------|----------------|----------------|----------------|----------------|----------------|
| 1 | A            | 2023-01-03              | AGI-LENT<br>TECH-NOLO-<br>GIES<br>INC | 11             | 3826         | 1414284        | 150.04         | 153.13         | 148.470        | 151.96         |
| 2 | A            | 2023-01-04              | AGI-LENT<br>TECH-NOLO-<br>GIES<br>INC | 11             | 3826         | 1247420        | 151.67         | 153.04         | 150.240        | 151.65         |
| 3 | A            | 2023-01-05              | AGI-LENT<br>TECH-NOLO-<br>GIES<br>INC | 11             | 3826         | 1714635        | 152.11         | 153.07         | 148.770        | 150.00         |



|   | tic<br><chr> | data-<br>date<br><date> | conm<br><chr>                                 | exchg<br><int> | sic<br><int> | cshtd<br><dbl> | prccd<br><dbl> | prchd<br><dbl> | prcld<br><dbl> | prcod<br><dbl> |
|---|--------------|-------------------------|-----------------------------------------------|----------------|--------------|----------------|----------------|----------------|----------------|----------------|
| 4 | A            | 2023-01-06              | AGI-<br>LENT<br>TECH-<br>NOLO-<br>GIES<br>INC | 11             | 3826         | 2444397        | 147.67         | 154.64         | 143.010        | 154.36         |
| 5 | A            | 2023-01-09              | AGI-<br>LENT<br>TECH-<br>NOLO-<br>GIES<br>INC | 11             | 3826         | 1269488        | 147.47         | 151.28         | 147.198        | 149.69         |
| 6 | A            | 2023-01-10              | AGI-<br>LENT<br>TECH-<br>NOLO-<br>GIES<br>INC | 11             | 3826         | 1565745        | 155.23         | 155.55         | 148.750        | 150.18         |

## Part II: Extended Analysis

Carry out the extended analysis in A2 Analysis.pdf to pick a specific security for further analysis

### 1. Calculate simple weekly returns for each ticker in the full dataset

```
# Calculating simple weekly returns for each ticker in the dataset
# Keep the last trading day in each ISO week per ticker
# Weeks start Monday
# Last Day Present
eow <- sp500 %>%
  mutate(week = floor_date(datadate, "week", week_start = 1)) %>%
  group_by(tic, week) %>%
  slice_max(order_by = datadate, n = 1, with_ties = FALSE) %>%
  ungroup() %>%
```

```

arrange(tic, week)

# Simple weekly returns: (P_t - P_{t-1}) / P_{t-1}
weekly_returns <- eow %>%
  group_by(tic) %>%
  mutate(R_weekly = (prccd - lag(prccd)) / lag(prccd)) %>%
  ungroup()

# Drop the first NA per ticker
weekly_returns <- weekly_returns %>% filter(!is.na(R_weekly))

head(sp500)

```

A data.frame: 6 × 11

|   | tic<br><chr> | data-<br>date<br><date> | conm<br><chr>                    | ex-<br>chg<br><int> | sic<br><int> | cshtd<br><dbl> | prccd<br><dbl> | prchd<br><dbl> | prcld<br><dbl> | prcod<br><dbl> | gvkey<br><int> |
|---|--------------|-------------------------|----------------------------------|---------------------|--------------|----------------|----------------|----------------|----------------|----------------|----------------|
| 1 | A            | 2023-01-03              | AGI-LENT<br>TECH-NOLOGIES<br>INC | 11                  | 3826         | 1414284        | 150.04         | 153.13         | 148.470        | 151.96         | 126554         |
| 2 | A            | 2023-01-04              | AGI-LENT<br>TECH-NOLOGIES<br>INC | 11                  | 3826         | 1247420        | 151.67         | 153.04         | 150.240        | 151.65         | 126554         |
| 3 | A            | 2023-01-05              | AGI-LENT<br>TECH-NOLOGIES<br>INC | 11                  | 3826         | 1714635        | 152.11         | 153.07         | 148.770        | 150.00         | 126554         |
| 4 | A            | 2023-01-06              | AGI-LENT<br>TECH-NOLOGIES<br>INC | 11                  | 3826         | 2444397        | 147.67         | 154.64         | 143.010        | 154.36         | 126554         |

|   | tic<br><chr> | data-<br>date<br><date> | conm<br><chr>                             | ex-<br>chg<br><int> | sic<br><int> | cshtd<br><dbl> | prccd<br><dbl> | prchd<br><dbl> | prcld<br><dbl> | prcod<br><dbl> | gvkey<br><int> |
|---|--------------|-------------------------|-------------------------------------------|---------------------|--------------|----------------|----------------|----------------|----------------|----------------|----------------|
| 5 | A            | 2023-01-09              | AGI-LENT<br>TECH-<br>NOLO-<br>GIES<br>INC | 11                  | 3826         | 1269488        | 147.47         | 151.28         | 147.198        | 149.69         | 126554         |
| 6 | A            | 2023-01-10              | AGI-LENT<br>TECH-<br>NOLO-<br>GIES<br>INC | 11                  | 3826         | 1565745        | 155.23         | 155.55         | 148.750        | 150.18         | 126554         |

## 2. Categorise your data into decile groups based on simple weekly returns;

Labelled 0%, 10%, 20%, ... Do not remove zero returns from the data

```
# Create decile groups based on simple weekly returns
weekly_returns <- weekly_returns %>%
  mutate(decile = ntile(R_weekly, 10)) # 10 groups = deciles
```

```
# Create percentage labels for deciles
decile_labels <- paste0(seq(0, 90, by = 10), "%")

weekly_returns <- weekly_returns %>%
  mutate(decile_label = factor(decile_labels[decile],
                               levels = decile_labels))
```

```
# Check

table(weekly_returns$decile_label)
summary(weekly_returns$R_weekly)
```

```
0% 10% 20% 30% 40% 50% 60% 70% 80% 90%
5202 5202 5202 5202 5202 5202 5201 5201 5201 5201
```

| Min.      | 1st Qu.   | Median   | Mean     | 3rd Qu.  | Max.     |
|-----------|-----------|----------|----------|----------|----------|
| -0.980486 | -0.019172 | 0.002282 | 0.002540 | 0.023447 | 0.784177 |

### 3. Display a table showing the top ticker in each decile group.

(The ticker with the highest weekly return)

```
# For each decile group, find the ticker with the highest weekly return
top_tickers <- weekly_returns %>%
  group_by(decile_label) %>%
  slice_max(order_by = R_weekly, n = 1, with_ties = FALSE) %>%
  ungroup() %>%
  select(decile_label, tic, datadate, R_weekly) %>%
  arrange(decile_label)

# Display the table
top_tickers
```

A tibble: 10 × 4

Table 12: Top ticker in each decile group

| decile_label <fct> | tic <chr> | datadate <date> | R_weekly <dbl> |
|--------------------|-----------|-----------------|----------------|
| 0%                 | DVN       | 2023-08-04      | -0.041509434   |
| 10%                | WBD       | 2023-05-19      | -0.024780176   |
| 20%                | CNC       | 2023-07-07      | -0.014084507   |
| 30%                | GLW       | 2024-12-31      | -0.005441607   |
| 40%                | BDX       | 2024-02-02      | 0.002281851    |
| 50%                | FE        | 2023-06-30      | 0.010132502    |
| 60%                | CNC       | 2024-01-12      | 0.018480227    |
| 70%                | TROW      | 2023-09-01      | 0.029170465    |
| 80%                | ETN       | 2023-02-10      | 0.046986033    |
| 90%                | SMCI      | 2024-11-22      | 0.784176534    |

### 4. Select the top ticker from the 60% decile group.

Use this ticker for the rest of the assignment, including in Part III.

```
# Select the top ticker from the 60% decile
top_60_ticker <- top_tickers %>%
  filter(decile_label == "60%")

top_60_ticker
```

A tibble: 1 × 4

Table 13: Top ticker from the 60% group

| decile_label <fct> | tic <chr> | datadate <date> | R_weekly <dbl> |
|--------------------|-----------|-----------------|----------------|
| 60%                | CNC       | 2024-01-12      | 0.01848023     |

```
# Use CNC (the top ticker from 60% decile)
selected_ticker <- "CNC"

ticker_data <- weekly_returns %>%
  filter(tic == selected_ticker) %>%
  select(-exchg, -cshtd) %>%
  arrange(datadate)

head(ticker_data)
```

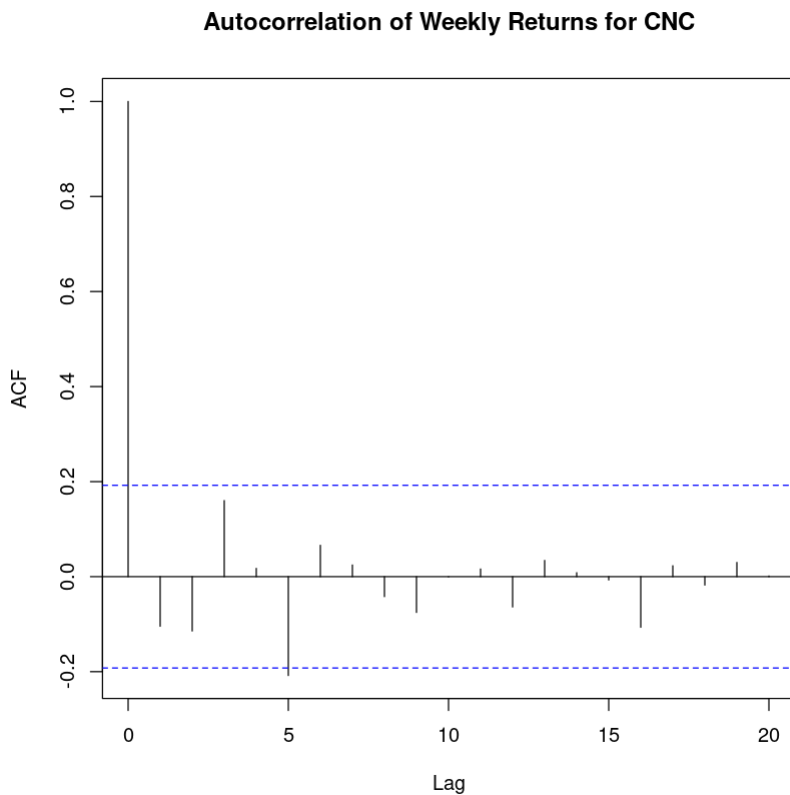
A tibble: 6 × 13

| tic<br><chr> | data-<br>date<br><date> | comm<br><chr>    | sic<br><int> | prccd<br><dbl> | prchd<br><dbl> | prcld<br><dbl> | prcod<br><dbl> | gvkey<br><int> | week<br><date> | R_weekly<br><dbl> | decile<br><int> | decile_la-<br>bel<br><fct> |
|--------------|-------------------------|------------------|--------------|----------------|----------------|----------------|----------------|----------------|----------------|-------------------|-----------------|----------------------------|
| CNC          | 2023-01-13              | CEN-TENE<br>CORP | 6324         | 77.04          | 79.065         | 76.870         | 77.73          | 145552         | 2023-01-09     | -0.020594966      | 3               | 20%                        |
| CNC          | 2023-01-20              | CEN-TENE<br>CORP | 6324         | 76.15          | 76.210         | 74.970         | 75.86          | 145552         | 2023-01-16     | -0.011552440      | 4               | 30%                        |
| CNC          | 2023-01-27              | CEN-TENE<br>CORP | 6324         | 75.25          | 76.830         | 74.895         | 76.63          | 145552         | 2023-01-23     | -0.011818779      | 4               | 30%                        |
| CNC          | 2023-02-03              | CEN-TENE<br>CORP | 6324         | 71.01          | 73.660         | 70.800         | 73.22          | 145552         | 2023-01-30     | -0.056345515      | 1               | 0%                         |

| tic   | data-<br>date | conm             | sic   | prccd | prchd  | prcld  | prcod | gvkey  | week       | R_weekly | decile | decile_la-<br>bel |
|-------|---------------|------------------|-------|-------|--------|--------|-------|--------|------------|----------|--------|-------------------|
| <chr> | <date>        | <chr>            | <int> | <dbl> | <dbl>  | <dbl>  | <dbl> | <int>  | <date>     | <dbl>    | <int>  | <fct>             |
| CNC   | 2023-02-10    | CEN-TENE<br>CORP | 6324  | 72.74 | 73.220 | 71.420 | 72.91 | 145552 | 2023-02-06 | 0.024362 | 2766   | 70%               |
| CNC   | 2023-02-17    | CEN-TENE<br>CORP | 6324  | 73.36 | 73.780 | 72.630 | 72.75 | 145552 | 2023-02-13 | 0.008528 | 2508   | 50%               |

## 5. Plot the autocorrelation function for this ticker's entire set of weekly returns.

```
# Base R autocorrelation plot
acf(ticker_data$R_weekly,
    main = paste("Autocorrelation of Weekly Returns for", selected_ticker),
    lag.max = 20) # 20 lags is typical for weekly data
```



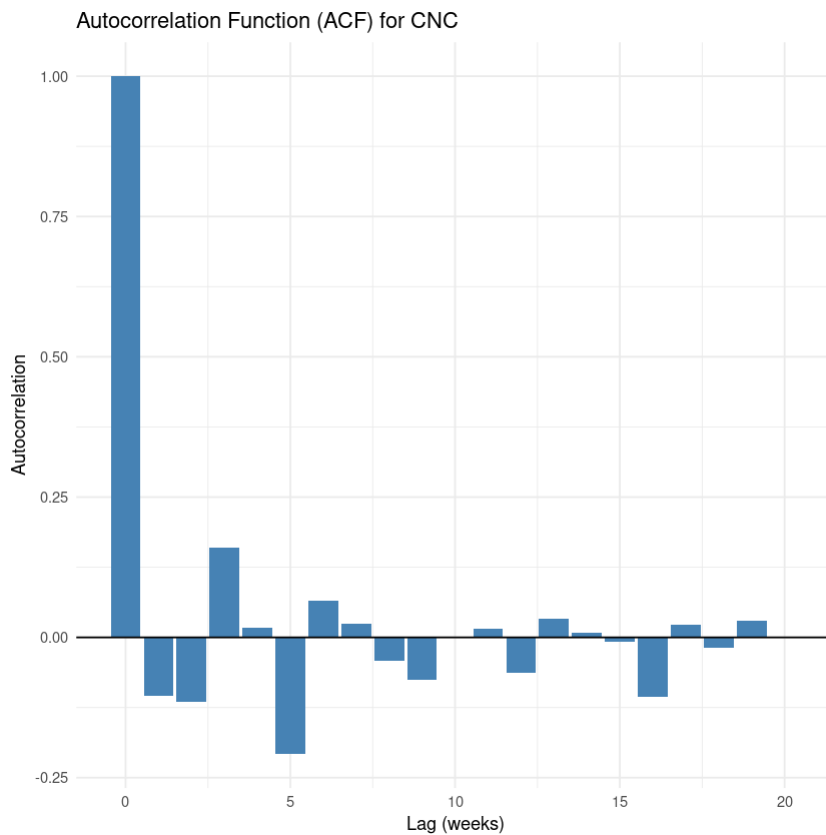
```

acf_values <- acf(ticker_data$R_weekly, lag.max = 20, plot = FALSE)

acf_data <- data.frame(
  lag = acf_values$lag,
  acf = acf_values$acf
)

ggplot(acf_data, aes(x = lag, y = acf)) +
  geom_bar(stat = "identity", fill = "steelblue") +
  geom_hline(yintercept = 0, color = "black") +
  labs(
    title = paste("Autocorrelation Function (ACF) for", selected_ticker),
    x = "Lag (weeks)",
    y = "Autocorrelation"
  ) +
  theme_minimal()

```



## Part III: Regression

```
# Part 3: Fama-French 3-Factor Regression

ff3 <- read.csv("fama_french_weekly.csv", skip = 4, check.names = FALSE)

# 1) Normalize header names so dashes etc. are safe to reference
names(ff3) <- gsub("-", ".", names(ff3))

# 2) First column is the YYYYMMDD date; rename and parse
ff3 <- ff3 %>%
  rename(yyyymmdd = 1) %>%          # rename the first column
  mutate(Date = as.Date(as.character(yyyymmdd), "%Y%m%d")) %>%
  filter(!is.na(Date))

# 3) Convert % -> decimals
ff3 <- ff3 %>%
  mutate(
    Mkt.RF = Mkt.RF/100,
    SMB     = SMB/100,
    HML     = HML/100,
    RF      = RF/100
  )

# 4) Align to your weekly key (ISO weeks start Monday)
ff3 <- ff3 %>% mutate(week = floor_date(Date, "week", week_start = 1))

# Merge with CNC's weekly returns
# ticker_data already has columns: week (Date), R_weekly (decimal)
merged <- ticker_data %>%
  select(week, R_weekly) %>%
  inner_join(ff3 %>% select(week, Mkt.RF, SMB, HML, RF), by = "week")

# 5) Excess return and FF3 regression
merged <- merged %>% mutate(Excess_Ri = R_weekly - RF)

ff3_model <- lm(Excess_Ri ~ Mkt.RF + SMB + HML, data = merged)
summary(ff3_model)
```

Call:



```
lm(formula = Excess_Ri ~ Mkt.RF + SMB + HML, data = merged)

Residuals:
      Min       1Q   Median       3Q      Max
-0.140978 -0.020295 -0.000082  0.023056  0.110701

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept) -0.003515   0.003809  -0.923   0.3584
Mkt.RF       0.369798   0.222869   1.659   0.1002
SMB          -0.287230   0.248363  -1.156   0.2502
HML          0.447264   0.223343   2.003   0.0479 *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.03772 on 100 degrees of freedom
Multiple R-squared:  0.05312,    Adjusted R-squared:  0.02472
F-statistic:  1.87 on 3 and 100 DF,  p-value: 0.1395
```

## Analysis

The Fama-French regression analysis for CNC, a moderately strong performer selected from the 60th decile, reveals a distinct, yet limited, systematic risk profile. The low market beta ( $\beta_{Market} = 0.37$ ) suggests defensive characteristics, implying CNC's returns offer relative stability and are less volatile than the overall market during broad swings. Crucially, the statistically significant positive exposure to the value factor ( $\beta_{HML} = 0.45$ ) indicates that CNC systematically benefits when value stocks outperform growth stocks. This links its risk/return profile to the value premium, a key consideration for portfolio managers seeking factor diversification. However, the very low  $R^2 \approx 0.053$  is the most vital finding. It signifies that over 94% of CNC's weekly return variation is unexplained idiosyncratic risk. This means firm-specific news, competitive developments, or regulatory changes, not market, size, or value movements are the primary drivers of its weekly performance. This suggests a deeper understanding requires analysis beyond these three factors, focusing on sector-specific (e.g., healthcare) drivers.